Recurrent-Access Debt as a Testable Constraint on Sequential Visual Report: A Staged Measurement-Validation and Causal Protocol

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Revision status and author response to the independent reviews

Both independent reviews identified the same central problem: the original manuscript assigned a mechanistic interpretation—feedback-attributable ATP-equivalent debt—to quantities that were not identified by the proposed measurements. We agree. In particular, magnetoencephalography, laminar-weighted functional MRI, arterial-spin labeling, oxygen concentration, lactate concentration, perfusion, and neural activity cannot simply be combined and relabeled as ATP expenditure or available reserve. A generic state-space model does not solve this problem unless its scales, structural zeros, temporal kernels, and sensor-specific forward relations are independently constrained.

The revised manuscript therefore makes five substantive withdrawals.

1. **The human predictor is no longer called trial-resolved ATP-equivalent expenditure.** Human imaging is assigned the more limited role of estimating condition-level, recurrent-associated metabolic proxies. No subsecond human metabolic recovery constant is claimed from hemodynamic data.

1. **The primary outcome no longer contains the recurrent neural phenomenon the hypothesis seeks to explain.** Correct delayed report is modeled through a choice or signal-detection likelihood. Confidence, metacognition, early target information, late target information, recurrent dynamics, and first-person visibility reports remain separate outcomes.

1. **The reduced logistic law is no longer presented as a mathematical consequence of the stochastic reserve equation.** The behavioral link is explicitly an additional modeling assumption. When debt is stochastic, the likelihood integrates over its posterior distribution. A flexible generic lag function is included so that average attentional-blink structure cannot automatically be attributed to metabolism.

1. **Raw expenditure and residual debt are no longer treated as interchangeable.** Finite-duration demand contributes through a recovery-weighted convolution and a separately calibrated depletion-retention coefficient. Time is referenced consistently to physical stimulus onset in the experiment and to an explicitly defined physiological reference in reduced models.

1. **The primate intervention is not represented as executable while its tool remains unspecified.** No confirmatory causal experiment should begin until the molecular or pharmacological target, delivery method, sensor calibration, dose, spatial spread, reversibility, injury criteria, and rescue controls are fixed and validated. Because the supplied records contain no such validation, this manuscript now presents a staged qualification pathway rather than pretending that the missing evidence has been supplied by revision.

We retain the term **recurrent-access debt** as the name of a theoretical latent variable. This retention is a limited point of disagreement with the strongest possible interpretation of the reviews: failure to measure a proposed variable at present does not make it incoherent as a theoretical construct. It does, however, prevent the construct from serving as a confirmatory empirical predictor until an identified observation model is demonstrated. Accordingly, all uses of “ATP-equivalent” below refer either to a theoretical scale or to a future calibrated invasive scale that must pass explicit validation gates. Human analyses remain in relative proxy units.

We also retain recurrent neural activity as a relevant mechanistic indicator. We agree that it must not define the behavioral outcome, but we do not infer that recurrence should be excluded from the study. It is instead modeled as a separate mediator, competitor, and physiological endpoint.

Finally, the revised falsification logic is modular. One adequately precise discordance between behavioral and physiological recovery constants counts against the shared-constant claim in that experiment. Failure is no longer required in both humans and macaques. Equivalence is evaluated jointly on the logarithm of recovery time rather than through overlap of separate uncertainty intervals.

Abstract

The attentional blink is a reduction in reporting a second target when it follows a first target within a restricted interval. One supplied record describes a deficit extending to approximately 600 ms and reports that conscious perception of a task-relevant first target was necessary but not sufficient to elicit the effect, with working-memory encoding offered as a possible mechanism [30]. Ongoing electrophysiological state has also been linked to attentional-blink access [27]. These observations motivate, but do not establish, a metabolic account.

This manuscript proposes **recurrent-access debt** as a falsifiable latent mechanism: a content-preserving recurrent episode associated with a first target may leave a transient local deficit in immediately available energetic support, reducing the probability or sensitivity of a later target-specific recurrent completion. The theoretical debt state obeys first-order recovery only as a deliberately simple candidate law. For finite-duration demand, residual debt at the second target is a recovery-weighted convolution of demand, not raw integrated expenditure. Conditional report likelihood is specified separately from the physiological dynamics, and stochastic debt is marginalized rather than replaced by its mean.

The revised human study does not claim direct measurement of trial-level ATP expenditure or subsecond reserve recovery. It tests whether randomized first-target conditions with independently estimated, condition-level recurrent-associated metabolic proxy differences show a cost-by-lag modulation of second-target report beyond a flexible generic lag function, sensory evidence, working-memory variables, recurrence, representational overlap, and arousal. Correct report, confidence, metacognitive efficiency, early target information, late target information, and subjective visibility are analyzed separately. Human laminar imaging is described as laminar weighted and feedback compatible, not as a measurement of apical dendritic metabolism.

A causal macaque study is conditional on prior validation. Sensor-specific forward models must distinguish neural demand, perfusion, oxygen delivery and consumption, lactate production and use, and any directly measured energy-state signal. An isolated-event protocol must recover a physiological time constant without using second-target behavior. A pathway-specific intervention must then demonstrably alter recovery while preserving early sensory encoding and baseline physiology. Only after those gates are met may suppression and rescue be tested as causal manipulations.

The proposal is rejected modularly if the human proxy-by-lag term has no reliable held-out effect; if a validated physiological recovery process is not single exponential; if an adequately precise physiological and behavioral time-constant comparison is discordant; if pathway specificity fails; or if verified recovery alteration does not produce the predicted behavioral change. The strongest conclusion available from a successful program would concern a local physiological constraint on sequential, confidence-sensitive visual report under near-threshold conditions. It would not establish a universal necessity for consciousness or identify metabolism with experience.

Keywords: attentional blink; sequential visual report; recurrent processing; metabolic constraint; energetic reserve; cortical feedback; laminar measurement; astroglia; glycogen; lactate; parameter recovery; measurement validation

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1. Scope, question, and claim hierarchy

1.1 Domain-limited research question

The revised research question is:

> **Under near-threshold two-target visual conditions, does a recurrent-associated local metabolic perturbation after T1 recover with a measurable time course that constrains later T2 report beyond generic lag effects and nonmetabolic bottlenecks?**

This question is narrower than the original formulation in four respects.

- It concerns **sequential visual report**, not consciousness in every modality or state.
- It concerns a probabilistic constraint, not a universal “only when” necessity.
- It distinguishes a theoretical debt state from the measurements used to estimate it.
- It requires validation before interpreting any latent component as ATP-equivalent debt.

The theory does not claim that one visual target measurably depletes total cortical ATP. Total ATP concentration can remain buffered while metabolic flux changes. Nor does it claim that all feedback activity is metabolically limited on the attentional-blink timescale. The central empirical question is whether there exists a local, behaviorally consequential recovery process with the predicted attribution, time course, and intervention sensitivity.

1.2 Claim hierarchy

The manuscript separates four levels of claim.

Level 1: broad motivation supported by the stored records

Information processing in the brain is energetically expensive, and cerebral metabolic activity is spatially heterogeneous [32]. Neural cell populations differ in metabolic capacities and exchange metabolites, linking bioenergetic organization to synaptic function and behavior [42]. A review title and metadata also support the broad statement that astroglia contribute to regulation of synapses, cognition, and behavior, although the supplied record does not permit detailed mechanistic verification [41].

Predictive-processing accounts describe cortical function through bidirectional exchanges between sensory evidence and top-down expectations [2]. A dreaming account similarly emphasizes internally generated models and discusses statistical and thermodynamic efficiency as part of embodied inference [1]. A conceptual criticality model associates conscious access with self-sustained propagation near a critical threshold [8]. These sources motivate investigation of recurrent dynamics, state dependence, and energetic constraints.

Level 2: inferential synthesis

If a task-relevant target recruits recurrent processing, and if some local support process recovers more slowly than the neural event itself, then the recent history of recurrent demand could alter the probability of later target completion. This inference is compatible with the broad records but is not demonstrated by them.

Level 3: named mechanistic hypothesis

The named hypothesis is that a first-target recurrent episode creates a **residual debt state** in pathway-specific feedback-recipient cortical compartments. The debt state is predicted to recover approximately exponentially over a defined operating range and to modulate later target-specific report.

Level 4: specific cellular implementation

The strongest proposed implementation attributes the rate-limiting support partly to astrocytic glycogen mobilization and predicts rescue by appropriately controlled local L-lactate delivery. The supplied records do not establish this implementation, its timescale, its selectivity, or its technical feasibility. It remains a contingent hypothesis that cannot enter confirmatory testing until separately validated.

1.3 What the program could establish

A successful program could support the following restricted conclusion:

> In a specified visual task, a pathway-specific local physiological recovery variable contributes to the lag-dependent probability or sensitivity of reporting a second target, after accounting for generic lag structure, early sensory encoding, working-memory variables, recurrent neural dynamics, arousal, and representational competition.

It could not establish that:

- failed T2 report implies absence of all phenomenal experience;
- metabolism constitutes consciousness;
- astrocytes or lactate carry representational content;
- every conscious event obeys the same recovery law;
- imagery is equivalent to dreaming;
- macaque confidence behavior is identical to human introspection;
- or a comparable resource process in an artificial system establishes machine consciousness.

1.4 Status as a staged program

The proposal has four stages.

1. **Stage 0: formal and simulation validation.** Establish mathematical coherence, parameter recoverability, and false-positive control.
2. **Stage A: measurement qualification.** Calibrate sensors and forward models, test isolated-event recovery, and determine whether a debt-like state is identifiable independently of behavior.
3. **Stage B: human predictive study.** Test condition-level proxy-by-lag associations without claiming trial-level ATP expenditure or a human physiological recovery constant.
4. **Stage C: invasive causal study.** Begin only after a named sensor and intervention package passes qualification.

Failure at an earlier stage prevents progression. This sequencing is not merely administrative. It prevents an underidentified latent decomposition from acquiring mechanistic credibility through a behavioral correlation.

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2. Theoretical background and evidential boundaries

2.1 Bidirectional inference and recurrent processing

Predictive-processing accounts characterize cortical processing as a hierarchical, bidirectional cascade in which top-down predictions interact with incoming signals and prediction errors [2]. This broad framework makes a purely feedforward account of perception incomplete, but it does not imply that every top-down signal is conscious or that feedback has a unique metabolic signature.

A related account describes the brain as an experience-shaped virtual-reality generator, especially visible in dreaming, and proposes that waking perception and dreaming differ in how generative models are constrained by sensory input [1]. That proposal motivates an internally generated transfer condition, but it supplies no evidence for a subsecond local reserve, an exponential recovery law, or equivalence between wakeful imagery and dreaming.

The recurrent-access-debt hypothesis therefore uses predictive inference only as functional context. The candidate cost-bearing event is not prediction error in general. It is a content-preserving recurrent episode that can be separately estimated from early sensory encoding. Surprise, cue validity, expected precision, and prediction error remain alternative explanatory variables.

2.2 Critical propagation as a competitor

A conceptual computational model proposes that conscious access corresponds to self-sustained percolation near a critical threshold and associates this regime with integration-related measures and a posterior information-sharing pattern [8]. Because the source is conceptual and computational, it does not establish that the proposed branching statistics are valid measurements of criticality in the present tasks.

Critical dynamics nonetheless provide a serious competitor. A prestimulus state closer to propagation threshold could produce stronger T1 recurrence, greater local metabolic demand, and weaker T2 processing through network competition. In that case, metabolism would correlate with the blink without causing it. Conversely, a local metabolic perturbation could reduce effective recurrent gain and thereby move a network away from critical propagation.

The revised design treats criticality estimates in three roles:

- as prestimulus predictors of T1 and T2 processing;
- as possible mediators of a metabolic manipulation;
- and as alternatives that may explain behavioral variation without a debt state.

No single statistic is treated as a definitive criticality measure. Results will be phrased as conditional on the validated estimators used.

2.3 Attentional blink and working-memory alternatives

The attentional blink is useful because target separation is experimentally controlled while T2 remains near report threshold. The supplied record by Ophir and colleagues describes a T2 report deficit within approximately 600 ms and reports that conscious perception of task-relevant T1, rather than allocation of spatial attention alone, was necessary but not sufficient to elicit the blink. The authors identify working-memory encoding as a possible mechanism [30].

This record directly constrains the current proposal. T1 working-memory encoding cannot be treated as a nuisance variable whose removal automatically reveals metabolism. It may be the primary bottleneck. The revised study therefore measures or manipulates:

- T1 task relevance;
- requirement to retain T1;
- consolidation interval;
- response order;
- representational overlap;
- and delayed-report demands.

A metabolic model must improve prediction beyond a working-memory or occupancy model. If it does not, the parsimonious conclusion is that measured metabolic responses track processing rather than impose an independent limitation.

Another supplied record links ongoing electrophysiological state to conscious access in the attentional blink [27]. Because only title-level information is available in the stored record, the manuscript does not infer a specific effect size, oscillatory mechanism, or causal direction. It supports only the broad requirement to measure ongoing state.

2.4 Recurrent and dendritic hypotheses

One title-level record links sustained attention, apical dendrite activity, and recurrent circuits [15]. The revised manuscript uses this only as broad theoretical motivation. It does not infer that apical dendrites are the measured metabolic compartment.

Anatomical terminology is kept explicit:

- **Layer I apical tufts** are subcellular compartments that can belong to pyramidal neurons with somata in deeper layers.
- **Supragranular populations** generally refer to neuronal populations in layers II and III.
- **Deep-layer populations** occupy layers V and VI.
- **Superficial and deep feedback-recipient zones** are pathway specific and are not synonymous with local recurrence.
- **Laminar-weighted fMRI** cannot resolve individual dendritic compartments or identify astrocyte-specific expenditure.

The human study therefore reports superficial, middle, and deep laminar-weighted signals with uncertainty. The invasive study must define feedback receipt for a named source-target pathway rather than assuming one generic “feedback layer.”

2.5 Metabolic heterogeneity without a proven debt mechanism

The brain’s energetic expense and regional heterogeneity are supported broadly by metabolic scaling work [32]. A more recent review describes differences among neural-cell metabolic profiles, intercellular exchange, and possible effects of bioenergetics on synaptic function and behavior [42]. These sources justify studying metabolic support but do not establish the following propositions:

- one successfully reported visual target produces a functionally limiting depletion;
- the depletion persists for roughly 100–600 ms;
- it is concentrated in feedback-recipient compartments;
- astrocytic glycogen is the rate-limiting process;
- local lactate restores the relevant variable selectively;
- or hemodynamic measurements can recover the proposed time constant.

Chronic associations among metabolic syndrome, insulin resistance, cognition, and neurodegeneration likewise do not validate an acute subsecond debt mechanism [43]. The timescales, pathologies, and causal structures differ.

2.6 Multidimensional evidence and consciousness terminology

An inside-out approach recommends convergence across behavioral, metacognitive, mnemonic, neural, and representational indicators rather than treating one measure as a decisive threshold for consciousness [3]. A bottom-up philosophy of animal consciousness similarly emphasizes an evolved, multidimensional phenomenon rather than an all-or-none human property [11].

The revised study adopts that pluralism while avoiding a new circularity. Convergence is evaluated across separate variables. It is not implemented by defining a positive outcome as the conjunction of correct report, high confidence, and recurrent neural activity. Such a conjunction would make recurrence part of the dependent variable and then claim that recurrence predicts the dependent variable.

Clinical work on disorders of consciousness further demonstrates that behavior, imaging, and electrophysiology can disagree and that ancillary techniques should not be elevated into infallible consciousness measures [23]. Recurrent-access debt is not proposed as a diagnostic marker.

2.7 Novelty boundary

Relative to the supplied records, none states the precise combination of:

- a finite-duration recurrent demand convolution;
- an independently estimated local recovery constant;
- a cost-by-lag modulation of sequential report;
- pathway-specific physiological qualification;
- and causal impairment plus parameter-specific rescue.

The proposal is therefore plausibly novel relative to this restricted source set. That is not a systematic novelty claim about the full literature. The supplied corpus omits substantial direct work on attentional-blink resource models, neurovascular coupling, cortical feedback anatomy, glycogen mobilization, fast metabolic sensors, and lactate pharmacology. Priority and feasibility would require a systematic search and direct technical literature before registration.

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3. Constructs, endpoints, and temporal references

3.1 Experimental events

For trial i and individual s:

- $T1$ is the first task-relevant target.
- $T2$ is the second task-relevant target.
- $t_{1,i}$ and $t_{2,i}$ are their physical onset times.
- The physical stimulus-onset asynchrony is

$$\Delta_i = t_{2,i} - t_{1,i}.$$

- Serial-position lag is recorded separately and is not assumed equivalent to milliseconds when stream rate varies.
- The onset, peak, and end of recurrent T1 activity are latent physiological times estimated from independent localizers. They do not replace physical Δ_i in task descriptions.

The confirmatory human task uses a fixed 100-ms item period. Candidate separations are 100, 200, 300, 400, 500, 600, 800, and 1,000 ms. The 100-ms condition is treated as a prespecified temporal-integration boundary condition, not as part of the primary single-exponential recovery test. The primary candidate recovery range is 200–800 ms. This range may be abandoned before registration if independent order-report calibration shows that 200-ms trials remain dominated by fusion. It cannot be altered after inspecting metabolic or debt-related results.

3.2 Primary behavioral endpoint

The primary endpoint is the participant's or animal's delayed T2 identity choice. It is defined independently of neural recurrence, metabolic measurements, and confidence.

For a balanced two-alternative target task, let:

- $S_{2,is} \in \{-1, +1\}$ denote true T2 category;
- $R_{2,is} \in \{-1, +1\}$ denote the reported category;
- $Y_{2,is} = 1$ if $R_{2,is} = S_{2,is}$, and 0 otherwise.

The confirmatory model uses the full choice likelihood rather than only percent correct, allowing perceptual sensitivity and choice criterion to be separated. Multicategory versions will use a multinomial evidence model with category-specific confusion parameters.

T1 report is modeled jointly rather than used as a selection criterion. The primary analysis does not restrict the dataset to $Y_1 = 1$, because conditioning on successful T1 report can induce selection bias when T1 state, recurrent processing, metabolic demand, and report share causes.

3.3 Confidence and metacognition

Confidence is recorded on an ordinal scale and analyzed continuously through an ordinal model. It is not thresholded to create a binary access label.

Separate estimates include:

- confidence calibration;
- type-2 sensitivity;
- metacognitive efficiency relative to type-1 sensitivity;
- response-specific confidence bias;
- and the effect of lag or intervention on confidence conditional on objective accuracy.

A high-confidence error remains informative. A correct low-confidence response is not automatically recoded as unconscious.

3.4 Neural endpoints

Neural outcomes are separated into at least four variables:

1. **Early T2 target information**, defined in an independently localized early sensory window.
2. **Late T2 target information**, defined in a later window without assuming recurrence.
3. **Feedback-compatible directed influence**, estimated using preregistered connectivity models.
4. **Duration or temporal generalization of target information**, measuring persistence.

These variables test the mechanistic prediction that debt affects later completion while sparing early encoding. They do not define behavioral success.

3.5 First-person reports

Human first-person outcomes include:

- visibility or clarity;

- temporal order;
- perceived fusion or separation;
- a report that "something appeared" without identification;
- confidence;
- and imagery vividness where applicable.

These reports are delayed and therefore memory dependent. They provide evidence about experience but cannot establish that an unreported target was phenomenally absent.

3.6 Recurrent-associated metabolic proxy

In the human study, $M_{R1,cs}$ denotes a **condition-level recurrent-associated metabolic proxy** for T1 condition c and participant s. It may combine calibrated oxygenation, perfusion, and laminar-weighted response features only after their sensor-specific meanings are retained in the model.

The proxy is not assumed to have ATP units. It is standardized within participant and acquisition context:

$$\widetilde{M}_{R1,cs} = \frac{M_{R1,cs} - \overline{M}_{R1,s}}{\mathrm{SD}(M_{R1,s})}, \tag{1}$$

where the mean and standard deviation are computed from calibration conditions without using T2 outcomes.

Human inference concerns differences among conditions, not absolute reserve depletion. Trial-level MEG can estimate recurrent neural variation but cannot convert an unobserved trial into a trial-specific metabolic history when imaging occurred in another session.

3.7 Theoretical energetic variables

In the physiological model:

- $E_R(t)$ is recurrent-associated energetic expenditure rate.
- $Q_{R1} = \int E_R(t)\,dt$ is raw integrated expenditure over a defined T1 interval.
- $D_R(t)$ is residual unavailable support, or debt, in the theoretical compartment.
- $B_R(t)$ is available support.
- B_0 is baseline available support.
- $D_R(t) = B_0 - B_R(t)$ only within the local operating range in which that subtraction is meaningful.
- ρ_R is the fraction or conversion coefficient linking expenditure to residual debt after accounting for concurrent supply.

The distinction between Q_{R1} and D_R is essential. A large flux need not produce a large residual deficit if supply rises simultaneously.

4. Causal structure and estimands

4.1 Proposed causal graph

The key variables are organized as follows:

```
Randomized T1 condition C1
        │
        ├──────────→ sensory evidence S1 ────→ early neural response F1
        │                                       │
        │                                       ▼
        ├──────────→ task relevance / WM demand W1 → recurrent response R1
        │                                                    │
        │                                                    ├──→ T1 report Y1
        │                                                    ├──→ metabolic demand E1
        │                                                    └──→ competition with T2
        │
Prestimulus state U → F1, R1, E1, T1 report, T2 processing, arousal measures
Representational overlap O12 → competition, T2 report
E1 + delivery P1 + buffering A1 → residual debt D1
D1 + generic lag processes G(Δ) + T2 evidence S2 → late T2 response R2 → T2 report Y2

Intervention Z → glycogen or support process → D1
Intervention Z may also directly affect baseline physiology, vascular state,
excitability, motivation, and neural gain.
```

The graph makes three cautions explicit.

First, matching or conditioning on R_1, confidence, or Y_1 can block part of the hypothesized pathway or open collider paths. Second, a local intervention can have several direct effects besides changing recovery. Third, metabolic proxy M_{R1} is a measurement of a downstream process, not a randomized exposure.

4.2 Human estimands

The human program distinguishes:

1. **Randomized total condition-by-lag effect**

$$\text{Effect}(C_1 \times \Delta \to Y_2),$$

estimated without conditioning on post-T1 mediators.

1. **Proxy moderation association**

$$\text{Association}(\widetilde{M}_{R1} \times \Delta, Y_2),$$

adjusted for prespecified pre-exposure confounders and compared with models containing recurrent dynamics and working-memory demand.

1. **Mechanistic indirect association**

$$C_1 \to R_1/E_1 \to Y_2,$$

treated as mediation only under explicit assumptions and sensitivity analyses.

The human metabolic result is predictive and associational. No available human manipulation is assumed to alter metabolic cost while leaving representational competition and task demand untouched.

4.3 Primate estimands

The invasive program distinguishes:

1. the total effect of a randomized intervention on physiological recovery;
2. the total effect on T2 report sensitivity;
3. the effect on early T2 neural information;
4. the relation between intervention-induced recovery change and behavioral change;
5. and an exploratory causal mediation effect.

Randomization identifies total treatment effects. It does not automatically identify mediation through recovery because recovery is post-treatment and can share unmeasured causes with behavior. Mediation requires negative controls, measured alternative pathways, and sensitivity analysis.

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5. Formal physiological model

5.1 Local reserve equation

The candidate local linear model is

$$dD_R(t) = \left[-\frac{D_R(t)}{\tau_R} + \rho_R E_R(t)\right] dt + \sigma_D\, dW(t), \tag{2}$$

where:

- $D_R(t)$ has debt units;
- $\tau_R > 0$ has units of time;
- $E_R(t)$ has energetic units per unit time;
- ρ_R has debt units per energetic unit;
- σ_D has debt units per square root of time;
- $W(t)$ is a standard Wiener process.

The equivalent reserve variable is $B_R(t) = B_0 - D_R(t)$.

Equation 2 is a local approximation. An additive Gaussian process is unbounded and can generate negative debt or reserve beyond a physical maximum. It is admissible only if fitted trajectories remain far from those boundaries. Stage 0 simulations require posterior boundary-crossing probability below 1% over the modeled interval. If that criterion fails, a bounded transformed state is required, for example

$$D_R(t) = D_{\max}\Lambda[\theta_R(t)]\,, \tag{3}$$

with dynamics defined on unconstrained $\theta_R(t)$. The simple exponential law is then only a small-signal approximation.

5.2 Finite-duration demand

Let t_a mark the start of the T1-associated demand interval and t_b a physiological reference time after that interval. Conditional on a deterministic expenditure trajectory and initial debt $D_R(t_a)$, the expected debt at t_b is

$$\mathbb{E}[D_R(t_b)] = D_R(t_a)\exp\left[-\frac{t_b - t_a}{\tau_R}\right] + \rho_R \int_{t_a}^{t_b} E_R(q)\exp\left[-\frac{t_b - q}{\tau_R}\right] dq. \tag{4}$$

The integration variable q has units of time. Equation 4 accounts for recovery while expenditure is occurring.

Define the residual T1 debt at reference time t_b as

$$C^D_{R1}(t_b) = \rho_R \int_{t_a}^{t_b} E_R(q)\exp\left[-\frac{t_b - q}{\tau_R}\right] dq. \tag{5}$$

This quantity is not equal to raw expenditure

$$Q_{R1} = \int_{t_a}^{t_b} E_R(q) dq \tag{6}$$

unless recovery during the event is negligible and ρ_R is known.

At the physical onset of T2, $t_{2,i}$, T1-associated debt is

$$D_{R1,i}(t_{2,i}) = \rho_R \int_{t_{a,i}}^{t_{2,i}} E_{R1,i}(q)\exp\left[-\frac{t_{2,i} - q}{\tau_R}\right] dq, \tag{7}$$

where $E_{R1,i}(q)$ excludes energetic demand attributed to T2 itself.

Equation 7 is the preferred mechanistic expression. It avoids treating physical onset separation as time since completion of a variable-duration process.

5.3 Pulse approximation

If T1 demand is short relative to τ_R and can be approximated as a pulse at $t_{p,i}$,

$$\rho_R E_{R1,i}(t) = C^D_{R1,i}\delta(t - t_{p,i}), \tag{8}$$

then for $t > t_{p,i}$,

$$D_{R1,i}(t) = C^D_{R1,i}\exp\left[-\frac{t - t_{p,i}}{\tau_R}\right]. \tag{9}$$

The delta pulse has debt units after integration; its instantaneous amplitude is distributional and should not be interpreted as an ordinary rate.

Equation 9 is a reduced approximation, not the primary definition of finite-duration cost. It may be used for visualization or parameter-recovery benchmarks if the approximation error is quantified.

5.4 Stochastic solution

Under Equation 2, the debt at T2 is random:

$$D_R(t_2) = D_R(t_a)e^{-(t_2-t_a)/\tau_R} + \rho_R \int_{t_a}^{t_2} E_R(q)e^{-(t_2-q)/\tau_R}dq + \sigma_D \int_{t_a}^{t_2} e^{-(t_2-q)/\tau_R}dW(q) \tag{10}$$

The sign of the stochastic integral can be absorbed into W; its variance is what matters. For constant τ_R, the conditional variance contributed by the process noise is

$$\mathrm{Var}[D_R(t_2) \mid E_R] = \frac{\sigma_D^2 \tau_R}{2}\left[1 - e^{-2(t_2-t_a)/\tau_R}\right], \tag{11}$$

before accounting for uncertainty in demand, loading parameters, and initial state.

The stochastic model does not reduce to a logistic model evaluated at expected debt. The outcome likelihood must integrate over $D_R(t_2)$, or the primary physiological model must set $\sigma_D = 0$ and place residual variation in the behavioral likelihood.

5.5 Repeated events

For pulse-like events j occurring at $t_j < t_n$, residual debt immediately before event n is

$$D_R(t_n^-) = D_{\mathrm{pre}}e^{-(t_n-t_0)/\tau_R} + \sum_{j<n} C_{Rj}^D e^{-(t_n-t_j)/\tau_R}. \tag{12}$$

This predicts history dependence. Preceding distractors, recent targets, and intertrial intervals must therefore be modeled or randomized. The two-target study is not isolated from prior trials unless sufficient recovery time is empirically demonstrated.

5.6 Single- versus multiple-timescale recovery

The named law assumes one dominant recovery constant over the tested range. Alternatives include

$$D_R(t) = \sum_{h=1}^{H} C_h e^{-t/\tau_h}, \tag{13}$$

where $H > 1$ indexes multiple recovery processes. A reliably superior multi-timescale model rejects the named single-exponential law. It does not confirm an unspecified “debt family”; any broader interpretation would require a new hypothesis.

6. Behavioral observation model

6.1 Generic lag and debt effects must be separate

An ordinary attentional-blink curve can arise without metabolic debt. The outcome model therefore contains a flexible generic lag function $g_s(\Delta_i)$ that captures temporal integration, overlap, working-memory occupancy, and other mean lag effects.

For a balanced binary T2 task, define latent type-1 sensitivity

$$d_{2,is} = d_{0,s} + g_s(\Delta_i) + \boldsymbol{\beta}^{\top}\mathbf{X}_{is} - \gamma_s D_{R1,is}(t_{2,i}), \tag{14}$$

where $\mathbf{X}_{is}$ contains prespecified predictors not already represented by g_s. The participant's choice follows

$$P(R_{2,is} = +1 \mid S_{2,is}, D_{R1,is}, \mathbf{X}_{is}) = \Lambda[c_s + S_{2,is}d_{2,is}], \tag{15}$$

where c_s is a response-criterion parameter. This coding allows debt to reduce discriminative sensitivity rather than simply shift response bias.

The logistic link is a modeling choice. It is not derived from metabolic recovery.

6.2 Marginal likelihood under stochastic debt

If debt is latent and stochastic, the observable likelihood is

$$P(R_{2,is} = +1 \mid S_{2,is}, \mathbf{X}_{is}, \mathcal{O}_i) = \int \Lambda\Big[c_s + S_{2,is}\{d_{0,s} + g_s(\Delta_i) + \boldsymbol{\beta}^{\top}\mathbf{X}_{is} - \gamma_s d\}\Big] p(d \mid \mathcal{O}_i)\, dd, \tag{16}$$

where $\mathcal{O}_i$ denotes physiological observations and calibration information. Equation 16 is generally a logistic–latent-state mixture, not a logistic model at mean debt.

A mean-debt approximation may be reported only if simulations show negligible bias over the observed variance range.

6.3 Human proxy model

Because human debt is not identified directly, the Stage B model uses the centered condition-level proxy from Equation 1:

$$d_{2,is} = d_{0,s} + g_s(\Delta_i) + \boldsymbol{\beta}^{\top}\mathbf{X}_{is} - \gamma_H \widetilde{M}_{R1,c(i),s} K(\Delta_i), \tag{17}$$

where $c(i)$ identifies the randomized T1 condition and $K(\Delta)$ is a prespecified recovery-shaped modulation.

If no independently validated human physiological time constant exists, K cannot be called a metabolic recovery kernel. The confirmatory options are then:

1. compare a prespecified exponential kernel with nonmetabolic shape alternatives using held-out prediction; or
2. use an externally validated invasive τ_R only after transportability criteria are met.

The ordinary lag curve is absorbed by g_s, while $\widetilde{M}_{R1}$ identifies differential modulation among conditions. Centering prevents the proxy term from automatically absorbing the average blink merely because all T1 events have positive cost.

6.4 Confidence model

Let $K_{2,is}$ be ordinal confidence. It is modeled using thresholds $\zeta_{s,r}$ on a latent confidence variable related to decision-evidence magnitude, response correctness, and possible metacognitive noise:

$$P(K_{2,is} \le r) = \Lambda[\zeta_{s,r} - \eta_{K,is}] \,. \tag{18}$$

Debt may predict confidence beyond type-1 evidence, but that effect is secondary. A confidence effect is not required to create the primary report endpoint.

6.5 Joint T1–T2 model

T1 and T2 choices are modeled jointly with shared prestimulus state U_i, condition effects, and correlated individual random effects. The primary analysis is unconditional with respect to observed T1 correctness.

A secondary model permits the T2 debt effect to vary with posterior evidence of T1 recurrent completion. This interaction tests the theoretical premise that a completed T1 episode produces more debt, but it does not select trials by observed T1 report. Any causal interpretation of this interaction is limited because recurrent completion is post-treatment.

6.6 Perturbation model

A metabolic intervention is not assumed to change only τ_R. Let intervention z affect:

- baseline support $B_{0,z}$;
- debt conversion $\rho_{R,z}$;
- expenditure amplitude $E_{R,z}$;
- recovery constant $\tau_{R,z}$;
- behavioral debt sensitivity γ_z;
- early sensory gain;
- generic lag function g_z;
- criterion c_z;
- and lapse rate.

The causal model compares these possibilities. A fitted increase in τ_R supports the proposed mechanism only if amplitude-only, baseline-only, vascular, and neural-gain alternatives perform worse and the sensors can identify the distinction.

The ordering $\tau_R^G > \tau_R^V$ does not by itself entail worse behavior under suppression. The behavioral prediction additionally requires comparable baseline support, debt amplitude, sensory evidence, and γ, or an explicit model of their changes.

7. Observation model and identifiability

7.1 Why a generic state-space model is insufficient

A latent vector labeled “feedforward,” “recurrent,” and “energetic” is not identified merely because those labels appear in an equation. With unrestricted transition and loading matrices, similarity transformations can rotate and rescale the latent coordinates while preserving the same observed likelihood.

There is also a direct scale confound:

$$D_R^\star = aD_R, \qquad \gamma^\star = \gamma/a,$$

which leaves γD_R unchanged. Absolute debt and debt sensitivity cannot both be inferred from behavior without an independent scale anchor.

The revised model therefore distinguishes a **forward-model template** from a validated measurement model. The template specifies what must be calibrated; it does not claim that calibration has already succeeded.

7.2 Physiological state vector

A candidate state vector for invasive validation is

$$\mathbf{x}(t) = \begin{bmatrix} F(t) \\ R(t) \\ E_F(t) \\ E_R(t) \\ P(t) \\ O(t) \\ L(t) \\ A(t) \\ D_R(t) \end{bmatrix}, \tag{19}$$

where:

- F is feedforward neural drive;
- R is recurrent or feedback-compatible neural drive;
- E_F and E_R are energetic demand associated with those neural processes;
- P is local perfusion or delivery state;
- O is tissue oxygen concentration or availability;
- L is lactate concentration;
- A is a directly measured energy-state or ATP-related quantity, if such a sensor is validated;
- D_R is the candidate residual debt state.

These states are not interchangeable. In particular, O, L, P, and A can change with different signs and delays depending on production, transport, consumption, buffering, and sensor kinetics.

7.3 Sensor-specific forward relations

A schematic neural observation model is

$$\mathbf{y}_{\text{neural}}(t) = \mathbf{H}_F F(t) + \mathbf{H}_R R(t) + \boldsymbol{\epsilon}_N(t). \tag{20}$$

For MEG, $\mathbf{H}_F$ and $\mathbf{H}_R$ include source geometry and temporal mixing. For laminar electrodes, they include depth-dependent sensitivity, volume conduction, and spike or field transformations.

A perfusion measurement is represented as

$$y_{\text{ASL}}(t) = h_P * P(t) + \epsilon_P(t), \tag{21}$$

where h_P is the modality-specific sampling and response kernel and $*$ denotes convolution.

A hemodynamic signal requires a vascular forward model:

$$y_{\text{BOLD}}(t) = h_B * \Phi[P(t), O(t), V(t)] + \epsilon_B(t), \tag{22}$$

where $V(t)$ denotes vascular volume and Φ maps physiological states to the observed signal. Raw BOLD is not ATP expenditure.

A tissue oxygen sensor obeys

$$y_O(t) = h_O * O(t) + \epsilon_O(t), \tag{23}$$

while oxygen dynamics may be schematized as

$$\frac{dO}{dt} = \text{delivery}[P(t)] - \text{consumption}[E_F(t), E_R(t)] - \text{clearance}[O(t)]. \tag{24}$$

A lactate sensor obeys

$$y_L(t) = h_L * L(t) + \epsilon_L(t), \tag{25}$$

with latent dynamics

$$\frac{dL}{dt} = \text{production} + \text{import} - \text{oxidation} - \text{export}\,. \tag{26}$$

If an ATP-related sensor is used,

$$y_A(t) = h_A * A(t) + \epsilon_A(t), \tag{27}$$

but its relation to debt must be calibrated. Total ATP concentration may remain stable while turnover changes, so $D_R \neq A_0 - A$ by definition.

7.4 Required identification constraints

A physiological debt model is admitted to confirmatory testing only if it includes all of the following constraints.

1. **Native sensor scale.** Each direct sensor remains anchored to its own calibrated physical units before any latent integration.
2. **Structural zeros.** For example, MEG has no direct loading on oxygen or lactate concentration; ASL has no direct loading on neural content.
3. **Fixed or independently estimated temporal kernels.** Sensor kinetics and hemodynamic kernels are not learned from T2 behavior.
4. **Sign constraints.** Consumption, delivery, and concentration relations are constrained according to independent calibration rather than chosen to maximize behavioral prediction.
5. **Separate delivery and use perturbations.** Calibration must include conditions that help distinguish increased supply from increased consumption.
6. **Neural attribution anchors.** Early feedforward and later feedback-compatible components are localized independently.
7. **A debt-state observation relation.** At least one measurement or experimentally validated combination must vary with remaining support rather than merely with expenditure.
8. **Absolute or invariant scale anchor.** ATP-equivalent terminology is permitted only if a stoichiometric or experimentally invariant conversion is established.
9. **No behavioral anchoring.** T2 choices, confidence, and blink curves are excluded from sensor calibration and recovery estimation.
10. **Held-out validation.** The fitted model must recover known simulated perturbations and predict independent experimental perturbations.

If no absolute scale anchor exists, the latent state is reported in standardized energetic-state units. Comparisons of γ across sessions, species, sensory conditions, or imagery conditions are then invalid unless measurement invariance is demonstrated.

7.5 Human observation model

Human MEG provides trial-resolved neural timing and content but no direct energetic measure. Human BOLD and ASL provide slow, indirect spatial and condition-level information. Constraining a hemodynamic model with MEG timing does not create subsecond metabolic information absent from the metabolic measurements.

Accordingly:

- human $M_{R1,cs}$ is estimated at the condition-by-participant level;
- human MEG recurrence is estimated at the trial level as a separate neural variable;
- imaging and MEG need not be treated as if they observed the same trial;
- a cross-session model assumes only condition-level exchangeability;
- and no human trial is assigned a metabolically measured $D_{R1,i}$.

Representational similarity analysis can relate content geometry across modalities without requiring one-to-one correspondence among sensors, voxels, or units [9]. It does not reconstruct the metabolic history of an unmeasured trial or establish causal direction.

7.6 Anatomical attribution

The candidate pathway for invasive qualification must be named before Stage A. One plausible visual implementation would examine feedback from an extrastriate source area to an earlier visual target area, but this manuscript does not designate a specific pathway as validated.

For each selected pathway, the protocol must separately define:

- layer I and superficial target zones;
- supragranular neuronal populations;
- deep feedback-recipient zones;
- middle-layer feedforward-dominant controls;
- local vascular compartments;
- and source-area activity.

Human laminar-weighted imaging can test whether proxy effects are stronger in superficial or deep bins than in middle bins. It cannot infer apical-tuft metabolism, cell type, or synapse direction from depth alone.

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8. Stage 0: simulation and parameter-recovery requirements

8.1 Purpose

Before data collection, simulations must determine whether the proposed parameters can be distinguished over the available lag, proxy, and noise ranges. A positive fitted γ is uninformative if generic lag, cost, τ_R, random effects, and measurement error trade off strongly.

8.2 Generative scenarios

Synthetic datasets will be generated under at least the following scenarios:

1. true single-exponential debt with known γ and τ_R;
2. $\gamma = 0$ with an ordinary attentional-blink lag curve;
3. recurrence-only impairment correlated with metabolic proxy;
4. working-memory occupancy correlated with proxy;
5. vascular-only transients with no energetic debt;
6. a two-exponential physiological recovery;
7. amplitude change without time-constant change;
8. baseline-support change without time-constant change;
9. intervention-induced early sensory loss;
10. cross-session instability of proxy scale;
11. latent-state rotations under insufficient loading constraints;
12. fusion or lag-1 sparing superimposed on later recovery;
13. heterogeneous animals with opposite intervention effects;
14. stochastic debt with enough variance to invalidate a mean-debt approximation.

Simulations will reproduce proposed sampling rates, sensor kernels, missingness, trial counts, participant counts, animal counts, layer misclassification, and realistic cross-session structure.

8.3 Recovery criteria

A model passes Stage 0 only if, across the preregistered parameter grid:

- nominal 95% intervals achieve at least 90% empirical coverage;
- median relative bias in τ_R and γ is below 20%;
- false-positive probability for a positive debt effect is at most 5% under each null scenario;
- the single-exponential model does not systematically defeat a true two-timescale model through regularization artifacts;
- posterior correlations do not make γ, τ_R, and generic lag effectively nonidentifiable;
- and held-out model comparison selects the correct model family with a prespecified minimum probability.

If these conditions fail at realistic sample sizes, the corresponding confirmatory test is removed rather than retained with broader rhetoric.

8.4 Nested prediction and leakage control

Every adaptive step must occur within training folds:

- preprocessing choice;
- source or layer selection;
- stimulus-condition selection;
- proxy construction;
- representational feature selection;
- state-space fitting;
- hyperparameter tuning;
- and lag-function complexity.

Expected log predictive density differences will be reported with standard errors or posterior uncertainty. A positive point estimate without uncertainty does not establish predictive superiority.

8.5 Power and sample size

Sample size will be set from the simulation grid after Stage A reliability is known. The original numerical recruitment targets are withdrawn as unsupported. A registration-ready protocol must specify:

- the minimum detectable standardized proxy-by-lag effect;
- expected proxy reliability;
- participant and trial attrition;
- the number of repeated imaging conditions needed for condition-level precision;
- and the minimum number of animals supporting population-level inference.

For invasive population claims, at least four animals with usable data in each within-subject intervention contrast are required. Trial count does not substitute for animals. If two or more animals show precise effects in the direction opposite to the group claim, pooled confirmation is withheld and heterogeneity becomes the primary result.

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9. Stage A: measurement and intervention qualification

9.1 Sensor qualification

Each proposed sensor must undergo bench or ex vivo characterization appropriate to its intended use. Qualification includes:

- response to steps and ramps in the target analyte;
- rise and decay constants;
- linear range and saturation;
- temperature dependence;
- pH sensitivity;
- oxygen sensitivity where relevant;
- cross-sensitivity to lactate, glucose, redox state, or motion;
- drift and photobleaching where relevant;
- spatial sampling volume;
- and calibration uncertainty.

A sensor whose own decay time overlaps the hypothesized physiological recovery cannot be used to infer τ_R without a validated deconvolution model. If deconvolution is unstable, the sensor fails qualification.

9.2 Recovery of known simulated perturbations

The complete forward model must first recover simulated perturbations not used for fitting. These perturbations vary independently in:

- neural demand amplitude;
- demand duration;
- perfusion;
- oxygen delivery;
- lactate concentration;
- baseline energetic state;
- recovery time;
- and sensor delay.

Successful recovery means identifying the correct sign, amplitude class, and time-constant class with calibrated uncertainty.

9.3 Isolated-event experimental protocol

A nonbehavioral or behavior-independent isolated-event protocol is required. It should produce a reproducible local neural demand event while varying interevent interval. The second event serves as a physiological probe of residual state, not as an access outcome.

The protocol must demonstrate:

1. reproducible local neural activation;
2. stable early neural input across intervals;
3. an interval-dependent physiological response beyond sensor refractory effects;
4. recovery estimates that generalize to held-out sessions;
5. stronger recovery evidence in the candidate pathway-specific compartment than in spatial controls;
6. and no use of T2 report to fit τ_R.

If no recovery process can be resolved over the 100–800-ms range, the attentional-blink-timescale debt hypothesis cannot proceed to causal testing. A slower recovery may be scientifically interesting but would not support the proposed law.

9.4 Experimental scale calibration

"ATP-equivalent" is admitted only if the combined measurements can be linked to a stable energetic scale through independently characterized conversion relations. The calibration must report uncertainty and demonstrate invariance across:

- sessions;
- baseline physiological states;
- intervention conditions;
- and relevant cortical compartments.

If the mapping changes after glycogen suppression or lactate delivery, a fixed ATP-equivalent loading is invalid. Results must remain in sensor-native or standardized latent units.

9.5 Intervention qualification

Before a causal experiment, the exact intervention must be named in a locked technical annex. The annex must specify:

- molecular or pharmacological target;
- formulation and vehicle;
- delivery apparatus;
- dose and concentration;
- infusion rate and duration;
- expected spatial spread;
- cell-type specificity assay;
- time to onset and washout;
- reversibility;
- tissue-injury assessment;
- effects on pH, osmolarity, temperature, and vascular tone;
- effects on baseline firing and evoked sensory responses;
- and stopping criteria.

The supplied source set provides no evidence that such a selective, reversible, cell-restricted intervention is already validated in macaque visual cortex. The manuscript therefore does not name an arbitrary tool. This is an unresolved feasibility limitation, not a detail to be filled after confirmatory data are collected.

9.6 Rescue qualification

L-lactate cannot be designated a specific rescue merely because it improves performance. Before Stage C, rescue must:

- restore the targeted physiological state in the isolated-event assay;
- not alter early sensory gain at the selected dose;
- not produce the same effect through matched pH or osmolarity changes;
- have a prespecified inactive or mechanistically distinct control validated in the same assay;
- and be evaluated for vascular, signaling, motivational, and excitability effects.

If no selective dose window exists, the rescue claim is not testable with that manipulation.

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10. Stage B: human predictive program

10.1 Design objective

The human program tests a modest question:

> Do randomized T1 conditions with different, independently estimated recurrent-associated metabolic proxies produce different T2 lag functions after generic lag, sensory, mnemonic, neural, and state variables are modeled?

It does not estimate absolute debt or a physiological recovery constant from BOLD or ASL.

10.2 Task

Participants complete a rapid serial visual presentation task with balanced binary T1 and T2 categories. The item period is fixed at 100 ms. T2 appears at candidate separations of 100, 200, 300, 400, 500, 600, 800, or 1,000 ms after T1 onset.

Each trial contains:

1. a variable prestimulus baseline;
2. the RSVP sequence;
3. T1 and T2;
4. delayed T1 and T2 identity prompts in counterbalanced order;
5. ordinal confidence ratings;
6. and intermittent visibility, order, and fusion reports.

T2 strength is adjusted in an independent calibration block so that long-lag performance is above chance and below ceiling. Staircase parameters are fixed before the confirmatory blocks.

The 100-ms condition is analyzed with an explicit temporal-integration component. It is not used to establish monotonic exponential recovery.

10.3 T1 condition construction

T1 conditions may vary in:

- task relevance;
- contextual predictability;
- representational complexity;
- memory requirement;
- masking;
- and inferential ambiguity.

Condition selection uses calibration data only. Conditions are not required to match post-treatment mediators such as recurrence or confidence in the primary total-effect analysis. Instead, the design seeks overlap in physical visibility and avoids floor or ceiling performance.

A separate associational analysis examines whether proxy differences predict behavior after recurrence and working-memory variables are included. That analysis is not interpreted as a total causal effect.

10.4 MEG session

MEG estimates:

- early T1 and T2 category information;
- later category information;
- temporal generalization;
- prestimulus state;
- feedback-compatible directed influence;
- response preparation;
- and candidate criticality statistics.

The early T2 window is fixed from a single-target localizer. The late window is also defined independently, using latency and connectivity criteria without reference to metabolic proxies or T2 outcomes.

Response mappings vary across blocks so category information can be distinguished from motor preparation. All decoding is cross-validated, and feature selection remains inside training folds.

10.5 Imaging session

A deeply phenotyped subset completes laminar-weighted 7-T fMRI and perfusion imaging. Because these measurements are slow, the design estimates condition-level response magnitude, spatial distribution, and recovery at long separations appropriate to the modality. It does not infer a 200–600-ms reserve trajectory from the hemodynamic response.

Cortical-surface registration is challenging because of folding geometry and intersubject variability [19]. Layer assignment therefore carries explicit uncertainty. Analyses report sensitivity to:

- superficial-vein exclusion;

- cortical curvature;
- point-spread assumptions;
- depth-bin definitions;
- registration error;
- and alternative vascular models.

Signals are described as superficial-, middle-, or deep-weighted. They are not called apical-dendritic measurements.

10.6 Proxy construction

The human proxy is built from imaging features that are independently associated with the T1 condition and with feedback-compatible neural timing. Sensor-native components remain visible in the report. A latent combination is used only if it shows:

- cross-session reliability;
- held-out condition prediction;
- stable loading signs;
- and invariance across the conditions being compared.

The proxy is constructed without T2 choices, T2 confidence, or the blink curve. If reliability is insufficient to rank conditions, the human metabolic analysis is declared inconclusive.

10.7 Representational alignment

Representational similarity analysis compares target geometry across MEG, imaging, behavior, and computational models. This approach was developed to relate representations across measurement modalities and model spaces without requiring direct unit correspondence [9].

A late signal counts as target specific only if it:

- generalizes to held-out exemplars;
- favors a target-content model over mask and response models;
- and survives removal of motor and confidence-related variance.

RSA supplies evidence of representational correspondence, not feedback direction or energetic expenditure.

10.8 Arousal and autonomic measurements

Pupil size, pupil derivative, eye position, cardiac activity, respiration, head motion, and blink events are recorded. A wearable-sensor review supports the broad usefulness of multimodal physiological measurements for stress-related states but does not validate them as complete measures of cortical arousal in this task [25].

These variables are therefore partial controls. Absence of an autonomic effect does not exclude all arousal mechanisms. Tonic and phasic components are modeled separately, and contaminated trials are flagged before outcome analysis.

10.9 Working-memory and decision controls

Separate blocks vary whether T1 must be retained. Response order is counterbalanced, and a postcue can occasionally indicate that only T2 will be reported. These manipulations estimate the contribution of:

- T1 consolidation;
- dual-report demands;
- decision interference;
- and report-order memory.

A debt effect that appears only when T1 must be retained may reflect working-memory encoding rather than a general recurrent-access liability.

10.10 Primary human test

The primary human test is the held-out contribution of

$$-\gamma_H \widetilde{M}_{R1,c(i),s} K(\Delta_i)$$

in Equation 17, beyond:

- a flexible generic lag function;
- T2 physical evidence;
- randomized T1 condition;
- prestimulus state;
- working-memory requirement;
- trial history;
- response order;
- and participant random effects.

A second model adds measured T1 recurrence, prediction error, representational overlap, and autonomic variables. The first estimates predictive modulation; the second asks whether the proxy contains information beyond those correlated processes.

Support requires both models to improve held-out prediction over their corresponding proxy-free alternatives. If the effect disappears after reliable recurrence or working-memory variables are included, the evidence favors those alternatives.

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11. Imagery transfer

11.1 Rationale and restricted interpretation

The imagery condition asks whether internally generated, content-specific visual processing produces a comparable proxy-by-lag pattern. Its motivation comes from accounts emphasizing internally generated models and virtual inference [1]. It is not a study of dreaming.

11.2 Procedure

Participants learn a set of visual objects or scenes. On imagery trials, a symbolic cue identifies the item to imagine during a fixed interval. T2 then appears at one of the prespecified separations.

Controls include:

- viewing the corresponding image;
- maintaining the symbolic cue without imagery;
- a matched nonvisual memory task;
- and imagery attempts later rated as weak or absent.

Imagery success is evaluated through self-report and content-specific neural geometry, but neither criterion defines T2 report.

11.3 Measurement invariance

A shared γ across sensory and imagery conditions is tested only if the metabolic-proxy scale is invariant. Required evidence includes:

- equivalent loading interpretation;
- stable proxy reliability;
- comparable sensor response functions;
- and no condition-specific rescaling needed to produce agreement.

Representational similarity alone does not establish metabolic scale invariance.

11.4 Modular outcomes

- A reliable imagery proxy-by-lag effect supports transfer to internally generated content.
- A precise absence rejects imagery transfer but not the sensory result.
- A different behavioral shape rejects a shared recovery form.
- Failure of scale invariance makes cross-condition γ comparison uninterpretable.

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12. Stage C: conditional invasive causal program

12.1 Entry criteria

Stage C begins only if Stage A demonstrates:

1. a reproducible local recovery signal on the relevant timescale;
2. an identified or adequately constrained physiological state model;
3. a named and qualified intervention;
4. a qualified rescue;
5. stable early sensory encoding under the selected intervention dose;
6. and successful simulation-based recovery of intervention effects.

Without these criteria, the invasive causal experiment is not ethically or inferentially justified.

12.2 Subjects and animal-level inference

The design is within subject and counterbalanced, but population claims require at least four animals with usable data for every primary intervention contrast. Each animal must contribute vehicle, suppression, rescue, and relevant control sessions.

Inference includes:

- animal-specific effects;
- hierarchical population effects;
- leave-one-animal-out prediction;
- and heterogeneity estimates.

A result driven by one animal is not confirmatory. If precise animal-level effects differ in direction, the group result is reported as heterogeneous rather than averaged into a single mechanistic claim.

12.3 Behavioral task

Macaques perform a delayed two-target discrimination matched as closely as feasible to the human temporal structure. Target categories, reward contingencies, and motor mappings are balanced.

Behavior is modeled through an explicit choice likelihood. Optional wagering or opt-out behavior is not treated as a direct confidence report. A type-2 model includes:

- perceptual evidence;
- reward magnitude;
- opt-out value;
- risk preference;
- motivational state;
- and response history.

A wager effect supports uncertainty-sensitive behavior, not direct access to phenomenal consciousness.

12.4 Pathway-specific localization

The causal target must be defined for a named source-target pathway. Localization combines:

- source-area and target-area activity;
- laminar current-source-density profiles;
- target-content decoding;
- timing;
- directional influence;
- and, where separately justified, causal source perturbation.

Superficial and deep target compartments are analyzed separately. Layer I apical-tuft zones are not equated with supragranular populations. Middle-layer and spatially adjacent regions serve as controls.

12.5 Neural measurements

Laminar recordings estimate:

- spiking;
- local field potentials;
- current-source-density proxies;
- early T2 category information;

- late T2 category information;
- source-target coupling;
- and local propagation statistics.

The early T2 endpoint is fixed from single-target and long-lag localizers. Intervention equivalence in early encoding is evaluated before interpreting later behavioral changes.

12.6 Physiological measurements

Oxygen, lactate, perfusion, and any energy-state signal are modeled in their native meanings. The primary physiological endpoint is not "ATP" unless the calibration gate has been passed.

Isolated T1 and paired-probe trials estimate:

- response amplitude;
- recovery constant;
- baseline state;
- and sensor-specific delays.

T2 behavioral trials are excluded from the physiological τ_R fit used for independent comparison.

12.7 Intervention conditions

After qualification, the counterbalanced conditions are:

1. vehicle;
2. metabolic-support suppression;
3. suppression plus qualified L-lactate rescue;
4. suppression plus the qualified inactive or mechanistically distinct control;
5. L-lactate alone;
6. a spatial-control infusion where ethically and technically feasible.

Infusion pH, osmolarity, volume, temperature, rate, and mechanical effects are matched. Session order is randomized, and intervention identity is blinded during acquisition and initial preprocessing.

12.8 Competing intervention models

The confirmatory analysis compares models in which the intervention changes:

- τ_R only;
- debt amplitude only;
- baseline support only;
- conversion coefficient ρ_R;
- neural demand E_R;
- vascular delivery;
- early sensory gain;
- recurrent gain;
- behavioral sensitivity γ;

- or combinations of these parameters.

A time-constant interpretation requires the τ_R-changing model to outperform amplitude and baseline alternatives on held-out isolated-event data.

12.9 Causal prediction

The strongest predicted pattern is:

1. suppression changes the independently measured recovery state;
2. the change is best characterized as a longer τ_R, not merely larger amplitude or slower sensor response;
3. early T2 sensory information remains equivalent;
4. short-lag T2 sensitivity decreases;
5. rescue restores the physiological recovery parameter;
6. rescue restores the short-lag behavioral curve;
7. inactive and spatial controls do not reproduce the chain.

A local intervention may have remote network effects. Work in rats on focal ischemia and cortical inactivation illustrates that local disturbances can alter distant activity and connectivity [26]. The analogy is limited, but it justifies measuring remote state rather than assuming functional locality from infusion location alone.

12.10 Mediation caution

Even with random intervention assignment, the statement

$$Z \to \tau_R \to Y_2$$

is not automatically identified as causal mediation. Unmeasured treatment effects may influence both recovery and report. The analysis therefore includes:

- measured early sensory and vascular pathways;
- remote-network variables;
- negative-control sites or compounds;
- sensitivity to unmeasured mediator–outcome confounding;
- and comparison of total with mediated effects.

Failure to identify mediation does not erase a randomized total effect, but it prevents the recovery parameter from being claimed as the unique mediator.

13. Statistical specification

13.1 Hierarchical structure

Human models include random effects for:

- participant;

- T1 and T2 stimulus identity;
- session;
- acquisition modality where applicable;
- and condition.

Primate models include random effects for:

- animal;
- session;
- electrode or sensor site;
- stimulus;
- and intervention period.

Trials are nested within sessions and individuals. Trial count is not treated as independent biological replication.

13.2 Generic lag function

The generic lag term $g_s(\Delta)$ is represented by a low-rank spline whose complexity is fixed or selected entirely within training folds. It may capture nonmonotonic structure, including residual lag-1 effects.

The debt or proxy kernel must improve prediction beyond this flexible baseline. This requirement prevents a positive cost term from merely recapitulating the average lag curve.

13.3 Priors

For the standardized human proxy effect,

$$\gamma_H \sim \mathrm{HalfNormal}(0, 0.5), \tag{28}$$

in sensitivity units per one within-participant proxy standard deviation at maximal kernel weight. Sensitivity analyses use scales of 0.25 and 1.0.

For a positive physiological recovery constant,

$$\log \tau_R \sim \mathrm{Uniform}\left[\log(0.05 \text{ s}), \log(2.0 \text{ s})\right], \tag{29}$$

unless Stage A establishes a narrower independent range. This is a proper bounded prior, not a claim that every value in the range is biologically equally plausible.

The point-null comparison sets $\gamma = 0$. Bayes factors are interpreted only with the stated alternative prior. Predictive criteria remain primary because Bayes factors can be prior sensitive.

13.4 Measurement error

Proxy uncertainty is propagated through the outcome model. If

$$M_{R1,cs} \sim p(M \mid \mathbf{y}_{cs}),$$

then the behavioral likelihood integrates over that distribution. Plug-in analyses using posterior means are secondary.

The same rule applies to:

- layer assignment;
- physiological τ_R;
- sensor calibration;
- and representational overlap.

13.5 Held-out evaluation

Three levels of prediction are required:

1. held-out trials within participant or animal;
2. held-out stimuli or conditions;
3. held-out individuals or leave-one-animal-out evaluation.

Reported metrics include:

- expected log predictive density and its uncertainty;
- calibration;
- Brier score for binary outcomes;
- confusion-matrix prediction for multicategory outcomes;
- and posterior predictive checks.

All data-driven condition selection and preprocessing choices occur within training folds.

13.6 Competing models

The prespecified competitors are:

1. **Generic-lag model:** flexible $g(\Delta)$ with no proxy or debt.
2. **Working-memory occupancy model:** T1 retention and consolidation variables.
3. **Recurrence-only model:** recurrent amplitude, duration, connectivity, and content persistence.
4. **Prediction-error model:** surprise, cue validity, and estimated prediction error, motivated by hierarchical predictive-processing accounts [2].
5. **Critical-state model:** prestimulus and post-T1 propagation statistics, motivated by the conceptual criticality account [8].
6. **Representational-competition model:** T1–T2 overlap and temporal interference.
7. **Amplitude-only physiological model.**
8. **Baseline-support model.**
9. **Single-exponential debt model.**
10. **Multi-timescale debt model.**

A generic metabolic interpretation is not allowed to survive every shape failure. The named single-exponential law fails if a prespecified alternative reliably predicts better.

13.7 Equivalence testing

Separate uncertainty-interval overlap is not used to test equality of recovery constants.

For a behavioral estimate τ_B and independently measured physiological estimate τ_P, define

$$\delta_\tau = \log \tau_B - \log \tau_P. \tag{30}$$

The proposed equivalence margin is

$$|\delta_\tau| < \log(1.25), \tag{31}$$

corresponding to a ratio between 0.80 and 1.25. On a recovery scale of several hundred milliseconds, a 25% difference approaches the duration of one RSVP item and is therefore considered mechanistically meaningful rather than negligible.

Equivalence requires either:

- a 90% interval entirely within the equivalence region under a frequentist two-one-sided-tests formulation; or
- posterior probability greater than 0.95 that Equation 31 holds.

A shared-constant claim is rejected in a given experiment if the 95% interval for δ_τ lies entirely outside the equivalence region. Otherwise the comparison is inconclusive.

One adequately precise discordant physiological experiment rejects the shared-constant claim for that domain. Agreement elsewhere cannot override it.

13.8 Early-decoding equivalence

“Preserved early T2 encoding” also requires equivalence. Its margin is set from baseline test–retest variability and the smallest early-decoding change that predicts a meaningful long-lag behavioral change. The value must be fixed before intervention labels are unblinded.

A nonsignificant early-decoding difference does not establish preservation.

13.9 Rescue criteria

Recovery is considered restored only if:

1. rescue differs from suppression in the predicted direction;
2. rescue is equivalent to vehicle on $\log \tau_R$;
3. early sensory measures remain equivalent to vehicle;
4. and rescue does not create broad lag-independent improvement.

13.10 Missing data and exclusions

Exclusions are based on acquisition quality, task compliance, or independent calibration—not on whether debt predictions are observed.

Lag-1 or fusion exclusions are determined from an independent calibration block and fixed physical timing. Trials are not excluded because neural patterns fail to separate after behavioral or metabolic outcomes are known.

Missing confidence or physiological data are modeled where plausible. Complete-case analysis is not the default if missingness relates to fatigue, motion, or performance.

14. Prespecified hypotheses and quantitative predictions

14.1 Human proxy-by-lag hypothesis

For standardized proxy difference $\Delta M > 0$, the proposed reduced modulation is

$$\Delta d_2(\Delta) = -\gamma_H \Delta M K(\Delta), \tag{32}$$

with $\gamma_H > 0$. Under an exponential candidate kernel,

$$K(\Delta) = e^{-\Delta/\tau_K}. \tag{33}$$

The effect must be strongest at shorter admissible separations and approach zero at long lag. It must improve held-out prediction beyond $g(\Delta)$.

14.2 Finite-duration physiological hypothesis

In invasive data, T2-onset debt is predicted by Equation 7 rather than unweighted integrated expenditure. A model using the recovery-weighted convolution should predict isolated-probe and behavioral outcomes better than one using Q_{R1} alone.

14.3 Pathway specificity

After matching measurement reliability, recovery-weighted demand in pathway-specific feedback-recipient compartments should predict later report more strongly than:

- middle-layer feedforward-dominant controls;
- spatially adjacent control sites;
- motor-related regions;
- or global physiological measures.

Failure of this contrast rejects layer or pathway specificity.

14.4 Early-versus-late dissociation

A valid debt manipulation should leave early T2 category information within the equivalence margin while reducing:

- late T2 category information;
- feedback-compatible coupling;
- or report sensitivity.

If early encoding changes enough to explain behavior, selective recurrent-access debt is not supported.

14.5 Perturbation predictions

The intended, but not guaranteed, parameter ordering is

$$\tau_R^G > \tau_R^V, \tag{34}$$

and

$$\tau_R^L \approx \tau_R^V, \tag{35}$$

where G denotes qualified support suppression, V vehicle, and L suppression plus qualified L-lactate rescue.

Behavioral impairment is predicted only after the full intervention model establishes that changes in baseline, amplitude, early sensory gain, criterion, and motivation do not account for the result.

14.6 Imagery hypothesis

If imagery produces a reliable, measurement-invariant recurrent-associated metabolic proxy, larger imagery proxy values should predict greater short-lag T2 impairment. Shared γ or shared shape is tested only after invariance. No transfer claim is made from neural representational similarity alone.

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15. Modular falsification and decision rules

15.1 Failure of human predictive association

The human proxy hypothesis is rejected if:

- the held-out proxy-by-lag effect is nonpositive with adequate precision;
- a Bayes factor of at least 10 favors the point null under the stated prior;
- or the proxy model fails to improve held-out prediction over the generic-lag model.

An in-sample positive coefficient is insufficient.

15.2 Failure of independent physiological recovery

The physiological debt hypothesis is rejected if a validated isolated-event assay shows no reproducible recovery signal in the relevant temporal range despite adequate sensor precision.

If a recovery signal exists only on a much slower scale, the attentional-blink-timescale version is rejected.

15.3 Shape failure

The single-exponential law is rejected if:

- a multi-timescale model;
- fixed-duration occupancy model;
- nonparametric lag interaction;
- or recurrence-only model

reliably outperforms it in held-out data.

The broader statement that “energy matters” is not treated as survival of the named law.

15.4 Recovery-constant discordance

A precise value of δ_τ outside the equivalence margin in any adequately validated experiment rejects the shared-constant claim in that domain. Human imprecision cannot rescue a discordant primate result, and primate agreement cannot rescue a discordant human result.

15.5 Failed causal chain

The causal debt claim is rejected in the invasive experiment if:

- suppression reliably changes physiological recovery but not T2 behavior;
- rescue restores recovery but not behavior;
- behavioral change does not covary with recovery after intervention;
- or behavior is fully explained by early sensory, vascular, motivational, or global-state changes.

15.6 Failure of intervention specificity

The proposed cellular implementation is rejected if:

- the intervention changes several physiological parameters without identifiable recovery selectivity;
- L-lactate alters early sensory gain;
- an inactive control reproduces rescue;
- rescue occurs without physiological restoration;
- or vehicle-matched pH, osmolarity, or vascular effects account for the result.

15.7 Failure of pathway specificity

The pathway-specific claim is rejected if control compartments predict or mediate behavior as strongly as the candidate feedback-recipient compartments after reliability and signal-to-noise differences are accounted for.

15.8 Failure of imagery transfer

A precise null or different shape in imagery rejects the imagery-transfer claim. It does not erase a separately supported sensory result.

15.9 Measurement failure

If the sensor model cannot distinguish expenditure, delivery, concentration, and residual support, no mechanistic behavioral test is interpreted. The result is a failed measurement program, not confirmation or falsification of an inaccessible latent variable.

16. Interpretation if the program succeeds

16.1 Strongest warranted conclusion

The strongest warranted conclusion would be that a local physiological recovery process contributes causally to sequential visual report under specified near-threshold conditions.

The interpretation would remain conditional on:

- the task;
- the selected pathway;
- the validated sensor model;
- the intervention;
- and the species studied.

It would not support a universal statement that representations become conscious only when a particular metabolic reserve is available.

16.2 Relation to recurrence

A successful early-versus-late dissociation would suggest that metabolism constrains the completion or maintenance of later target-specific processing rather than initial sensory registration. It would not show that recurrence is sufficient for experience.

If recurrence fully mediates the behavioral effect, metabolism may be interpreted as support for recurrent processing. If metabolism retains predictive value after reliable recurrence measures, it may identify a support variable not captured by neural amplitude or duration alone.

16.3 Relation to criticality

If recovery changes critical-state estimates, and those estimates mediate report, the result would support a metabolism-to-network-state pathway. If criticality explains all effects and debt adds no information, a network-state account would be favored.

Because the supplied criticality source is conceptual, any empirical claim must remain tied to the specific propagation statistics validated in the experiment [8].

16.4 Relation to working memory

If effects occur only when T1 must be retained, the result may concern working-memory encoding rather than perceptual access broadly. This would align with the working-memory interpretation raised in the supplied attentional-blink record [30].

A truly broader recurrent-access constraint should also appear when T1 is behaviorally relevant but does not require prolonged retention, provided recurrent completion still occurs.

16.5 Fixed capacity as a possible limiting approximation

A physiological recovery process can resemble fixed capacity when τ_R is long relative to tested lags. Conversely, rapid recovery can make a bottleneck difficult to detect. This provides a possible mechanistic reinterpretation of one form of serial limitation without denying other bottlenecks.

The proposal does not imply that all conscious processing is serial. Multiple features may be integrated within a scene while successive reportable target episodes still interfere.

17. First-person and philosophical implications

17.1 Report is evidence, not identity

Correct delayed report depends on perception, memory, decision, criterion, metacognition, and motor execution. Failed report therefore does not prove absence of phenomenal experience.

The revised program separates:

- target choice;
- confidence;
- subjective visibility;
- temporal-order experience;
- early neural information;
- late neural information;
- and physiological recovery.

Convergence strengthens interpretation, but disagreement is scientifically meaningful rather than something to be erased by a composite label.

17.2 Limited philosophical formulation

The original universal formulation is withdrawn. The revised claim is:

> Under some near-threshold sequential visual conditions, recent recurrent processing may temporarily reduce the physiological support available for another content-preserving recurrent episode, thereby lowering the probability or sensitivity of later confidence-sensitive report.

This is compatible with a dynamic view of conscious time in which contents have nonzero duration and different processes operate on different temporal scales [6]. The supplied synaptic-clock account is theoretical, and its time constants are not assumed to equal the proposed recovery constant.

17.3 What debt does not explain

The model does not explain:

- qualitative character;
- first-person perspective;
- why neural representation is accompanied by experience;
- unity of consciousness;
- or the normative significance of conscious states.

An inside-out approach emphasizes qualitative richness, situatedness, interpretation, integration, and dynamic stability [3]. Recurrent-access debt addresses only a possible temporal constraint on one operational form of access.

17.4 Distant alternatives

Accounts based on dynamically multivalued complexity [5], quantum processes in microtubules [10], or behavioral and social organization [4] are not directly tested by this program. Confirmation would identify a mesoscopic physiological constraint, not settle the ontology or function of consciousness.

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18. Limitations and unresolved objections

18.1 No direct feasibility evidence is supplied

The most important limitation remains unresolved: this manuscript contains no pilot demonstration that a local energetic reserve can be measured with the required temporal and spatial precision. The proposed validation gates are not substitutes for evidence.

Until such evidence exists:

- D_R is a theoretical state;
- human M_{R1} is a relative proxy;
- human subsecond τ_R is not operational;
- and the primate causal experiment is conditional.

18.2 Feedback-specific expenditure may remain unidentifiable

Even invasive neural and metabolic measurements may not uniquely partition energy use among:

- feedforward synapses;
- feedback synapses;
- local recurrence;
- neurons;
- astrocytes;
- vascular cells;
- transmitter recycling;
- and ion-gradient restoration.

A well-constrained model may identify a pathway-associated energetic state without identifying expenditure at individual feedback synapses. The claim must follow the achieved resolution.

18.3 A debt state may not correspond to a single physical pool

“Debt” can summarize delayed restoration across several coupled processes. If oxygen, lactate, redox state, ion gradients, and vascular delivery recover at different rates, one state may be inadequate.

A composite state can still be predictive, but it should not be described as one molecular reserve. Multiple-timescale recovery would reject the named single-exponential form.

18.4 Human hemodynamics cannot establish subsecond recovery

Human BOLD and perfusion measurements are slow and indirect. MEG timing does not convert them into fast metabolic measurements. The human program therefore cannot independently establish a 200–600-ms metabolic recovery constant with the proposed modalities.

If no direct fast human assay becomes available, physiological τ_R testing must remain invasive. Cross-species transfer would then require explicit transportability assumptions.

18.5 Cross-session proxy stability is uncertain

When MEG and imaging occur in different sessions, the human proxy is condition level. It assumes that the relative metabolic ordering of conditions is stable across sessions. Fatigue, learning, vascular state, and strategy may violate this assumption.

Retest reliability and condition-rank stability are required. Trial-level language is prohibited for unobserved imaging trials.

18.6 Scale invariance may fail

Without an absolute scale, γ is expressed per standardized proxy unit. It cannot be compared directly across:

- participants;
- sessions;
- sensory and imagery conditions;
- humans and macaques;
- or intervention states

unless measurement invariance is established. Similar behavioral coefficients on different scales do not demonstrate a common biological sensitivity.

18.7 A flexible generic lag term can reduce power

Including $g(\Delta)$ is necessary for a fair test, but it competes with the proposed kernel. If proxy variation is small or strongly tied to condition, γ and the generic lag interaction may remain weakly identified.

That outcome is a design limitation, not grounds for removing the generic lag term.

18.8 Recurrent neural measures remain model dependent

Late timing, target information, directed connectivity, and laminar profile can support a feedback-compatible interpretation, but they do not prove anatomical recurrence. Working memory, common input, and local persistence can produce similar signals.

The study must distinguish “feedback compatible” from “directly observed feedback.”

18.9 Anatomical resolution is limited

Human laminar imaging is influenced by draining veins, cortical curvature, point spread, registration, and depth-dependent coupling. It cannot resolve individual apical dendrites or astrocyte-specific metabolism.

Even invasive depth recordings sample local populations imperfectly. Pathway attribution requires convergent anatomy and physiology.

18.10 T1 conditioning and collider bias

Restricting analysis to correctly reported T1 trials can induce selection bias. The joint model reduces this problem but does not identify every latent cause of T1 and T2 report.

Randomized T1 manipulations carry more causal weight than post hoc trial strata.

18.11 Human condition manipulations may not isolate cost

Changes in complexity, predictability, task relevance, or memory demand can affect both metabolic proxy and cognitive competition. No proposed human manipulation is known to vary metabolic cost alone.

Human evidence therefore remains predictive. The causal burden lies with a qualified invasive intervention.

18.12 Fusion and lag-1 sparing complicate the law

At very short intervals, T1 and T2 may be integrated, confused in order, or share a processing episode. A supplied computational preprint reports simulation of lag-1 sparing and later attentional-blink patterns without the proposed metabolic law [33]. Its evidential weight for human mechanism is limited, but it illustrates that nonmonotonic behavior is possible.

The primary exponential test excludes the independently defined integration regime. If nonmonotonicity remains beyond that regime, the single-exponential law fails.

18.13 Intervention selectivity may be unattainable

Suppressing glycogen mobilization could affect:

- baseline support;
- excitability;
- transmitter handling;
- ion homeostasis;
- vascular coupling;
- motivation;
- and recurrent gain.

L-lactate may act through metabolic, signaling, or vascular pathways. The required selective dose window may not exist. If so, the specific glycogen–lactate implementation cannot be tested by the proposed design.

18.14 Confidence and wagering are not direct consciousness measures

Human confidence is influenced by criterion and metacognition. Macaque wagering is influenced by reward, risk, learning, and motivation. Explicit models reduce but do not eliminate these ambiguities.

No species comparison should treat the two response forms as phenomenologically identical.

18.15 Causal mediation remains assumption dependent

Randomization of the intervention does not randomize the mediator. Recovery–behavior mediation may remain confounded by unmeasured treatment effects. The strongest defensible conclusion may be that intervention changes both recovery and behavior in a coordinated way, not that recovery uniquely mediates the effect.

18.16 Novelty and literature coverage remain provisional

The supplied source set supports broad motivation but not direct feasibility. Before registration, the literature review must be expanded systematically to include direct evidence on:

- attentional-blink bottlenecks;
- neurovascular and neurometabolic coupling;
- pathway-specific feedback anatomy;
- cortical glycogen dynamics;
- fast oxygen, lactate, redox, and ATP-related sensors;
- and lactate intervention pharmacology.

The absence of the exact law from the supplied records is not sufficient to establish global novelty.

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19. Ethics

19.1 Human research

Human procedures involve near-threshold visual stimulation, MEG, high-field MRI, physiological monitoring, and introspective reporting. Requirements include:

- screening for MRI contraindications;
- visual sensitivity screening;
- fatigue limits;
- scheduled breaks;
- stopping criteria;
- and withdrawal without penalty.

No metabolic drug manipulation is proposed in humans. First-person responses are treated as potentially sensitive data.

Fatigue is both an ethical issue and a confound. Session length must be determined from independent tolerability testing rather than maximized for trial yield.

19.2 Nonhuman primate research

The invasive study is justified only if:

- Stage A demonstrates measurement feasibility;
- the causal question cannot be answered noninvasively;
- the intervention has a realistic prospect of mechanistic discrimination;
- and animal numbers are sufficient for biological replication without avoidable excess.

Refinement includes:

- positive-reinforcement training;
- minimal necessary restraint;

- veterinary oversight;
- analgesia and perioperative care;
- conservative infusion limits;
- injury monitoring;
- predefined humane stopping rules;
- and post-study care consistent with institutional requirements.

A failed behavioral report is not presumed to indicate suffering, but lack of verbal report is not presumed to exclude experience. Multidimensional approaches to animal consciousness support caution under uncertainty [3] [11].

19.3 Clinical nonapplication

The model is not a clinical diagnostic. Disorders-of-consciousness assessment is vulnerable to behavioral misdiagnosis, and imaging or electrophysiology serves as ancillary evidence rather than an independent oracle [23].

No patient's consciousness, prognosis, or treatment should be inferred from metabolic efficiency or a fitted recovery constant without dedicated prospective validation.

19.4 Artificial systems

Recurrent artificial models can show attention-like behavioral signatures and systematic perturbation responses [46]. Multimodal models can also perform poorly on visual-perception benchmarks despite strong linguistic abilities [34]. Neither result establishes consciousness.

A biological debt law cannot be transferred to machines by analogy with electricity, recurrence, memory limits, or heat. An inside-out assessment would require convergence across flexible behavior, internal modeling, memory, metacognition, and other indicators [3]. A philosophical proposal has argued that governance under uncertainty may focus on states that would be valenced if conscious [7], but that proposal is normative rather than empirical.

19.5 Dual-use concerns

If a local recovery process were validated, attempts might be made to enhance sustained attention or report pharmacologically. Potential risks include:

- coercive productivity enhancement;
- unequal access;
- unsafe metabolic manipulation;
- chronic vascular or metabolic effects;
- and use in high-risk occupational settings without adequate consent.

Initial translation should prioritize restoration of impaired function, not enhancement of healthy individuals.

20. Open-science and reporting plan

A registration-ready version must include the complete technical annexes omitted here. Before outcome labels are unmasked, the following must be public or time-stamped:

- exact task code;
- stimulus set and randomization;
- sensor identities and calibration;
- intervention identity and dose;
- anatomical pathway definition;
- sample sizes and attrition rules;
- lag range;
- fusion criteria;
- preprocessing pipelines;
- priors;
- equivalence margins;
- model formulas;
- simulation code;
- power results;
- and all falsification rules.

The final report must provide:

- sensor-native data alongside latent estimates;
- animal-level effects;
- participant-level proxy reliability;
- calibration failures;
- all negative controls;
- uncertainty in ELPD differences;
- posterior predictive checks;
- model-comparison sensitivity to priors;
- and clear separation of registered from exploratory analyses.

Null results are interpretable only when measurement qualification succeeds. If qualification fails, that failure must be reported rather than hidden by proceeding to an underpowered behavioral analysis.

21. Expected outcome classes

21.1 Measurement success and mechanistic success

The strongest outcome would combine:

- identified physiological recovery;

- a valid single-exponential approximation;
- positive held-out proxy-by-lag prediction in humans;
- pathway specificity;
- preserved early T2 information;
- intervention-induced recovery change;
- corresponding behavioral change;
- and qualified rescue.

This would support a local physiological constraint on sequential report.

21.2 Measurement success but behavioral failure

If recovery is measured and manipulated but behavior does not change, the proposed causal debt mechanism is rejected. The physiological process may exist without limiting performance in the tested range.

21.3 Behavioral association but measurement failure

If a human proxy predicts the blink but no debt state can be independently recovered, the result supports only a condition-level association. It cannot be called ATP-equivalent debt.

21.4 Recurrence-only result

If recurrent strength, duration, or connectivity explains held-out behavior and the metabolic proxy adds nothing, recurrence-only or working-memory accounts are favored.

21.5 Critical-state result

If propagation statistics explain physiological and behavioral changes without residual debt contribution, a network-state account is favored, subject to the validity of those statistics.

21.6 Multi-timescale result

If physiology and behavior require multiple recovery constants, the single-exponential law is rejected. A revised model would require independent registration rather than post hoc relabeling.

21.7 Sensory-specific but not imagery-general result

A sensory effect with a precise imagery null narrows the hypothesis. It may indicate dependence on sensory-driven competition, pathway recruitment, or a sensory metabolic process rather than a general recurrent-inference debt.

21.8 Inconclusive result

The result is inconclusive if:

- proxy reliability is low;
- physiological sensors cannot resolve the relevant interval;
- intervention effects on recovery are absent;
- early sensory equivalence cannot be established;
- or the data cannot distinguish lag and cost terms.

“Inconclusive” is not evidence for the debt hypothesis.

22. Conclusion

Recurrent-access debt is retained as a theoretical proposal but substantially narrowed as an empirical claim. The central candidate mechanism is a transient local deficit in immediately available support following recurrent-associated T1 processing. For finite-duration demand, the relevant variable is a recovery-weighted debt state,

$$D_{R1}(t_2) = \rho_R \int_{t_a}^{t_2} E_{R1}(q) \exp\left[-\frac{t_2 - q}{\tau_R}\right] dq,$$

not raw integrated expenditure. The behavioral likelihood contains a flexible generic lag term and treats the debt effect as an additional link-model assumption. When debt is stochastic, its uncertainty is integrated rather than collapsed into an expected value.

The human study can presently test only whether a condition-level recurrent-associated metabolic proxy modulates T2 report across lag beyond ordinary blink structure, working-memory demands, neural recurrence, representational overlap, sensory evidence, and measured state. It cannot recover trial-level ATP expenditure or an independent subsecond metabolic time constant from MEG, BOLD, and perfusion.

The stronger causal claim belongs to an invasive program that has not yet met its feasibility requirements. Sensor-specific forward models, isolated-event recovery, scale calibration, pathway localization, intervention specificity, and rescue pharmacology must all be demonstrated before confirmatory testing. The absence of those demonstrations is an unresolved limitation rather than a problem that mathematical notation can solve.

The theory is deliberately vulnerable. A precise null proxy effect rejects the human predictive claim. A nonexponential physiological trajectory rejects the named recovery law. One adequately precise physiological–behavioral mismatch rejects the shared-constant claim in that experiment. A verified recovery change without the predicted behavioral change rejects the causal debt interpretation. Loss of pathway specificity or preservation failure in early sensory encoding favors alternative mechanisms.

If the full sequence succeeds, it would identify a local physiological recovery process that constrains sequential, confidence-sensitive visual report under specified conditions. It would not establish that metabolism is consciousness, that failed report proves phenomenal absence, or that one cellular substrate universally governs conscious access. The intended contribution is narrower: to convert a speculative resource metaphor into a staged measurement problem with explicit scale constraints, independent validation, and outcomes capable of proving the proposal wrong.

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Peer Review Notes

- keiko-arendt: The manuscript presents a genuinely testable and potentially novel metabolic-recovery hypothesis, and the deterministic impulse-response derivation is largely correct under its stated simplifications. However, the central quantity, feedback-attributable ATP-equivalent debt, is not identifiable from the proposed human measurements, the full stochastic model does not yield the stated logistic law without additional approximations, finite-duration expenditure is defined inconsistently, and the access outcome circularly includes the recurrent neural phenomenon the model is meant to explain. Direct feasibility evidence and a revised observation model are required before this can function as a registered protocol.
- marisol-quade: The manuscript is conceptually original, unusually transparent about its speculative status, and commendably falsifiable. Its deterministic reserve-recovery and logistic equations are mostly coherent, and it repeatedly avoids identifying metabolism, recurrence, report, or consciousness. However, the central quantities—trial-level feedback-attributable ATP-equivalent cost and an independently measured subsecond recovery constant—are not identifiable from the proposed human measurements or from the stated generic state-space model. The stochastic reserve equation also does not marginalize to the asserted logistic law, the primate manipulation remains an unspecified placeholder, and the operational outcome partly defines access by the recurrent neural event the theory seeks to explain. The supplied sources support the broad ingredients but not the layer-specific glycogen–lactate mechanism, timescale, measurement feasibility, or novelty claim.